

Problem Set 2

Lauren Rackley

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1.

a) The response variable is dichotomous. In this case, the response variable is favorable or unfavorable opinion response to stress and residence. It is assumed that these data arise from a stratified simple random sample so that the favorable v. unfavorable response is binomially distributed for each stress $\times$ residence combination.

The regression equation is:

$$\text{logit}(\theta_{ij}) = \alpha + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3$$

where θ_{ij} = probability that person of i th type of residence with j th stress level has favorable response for $i = 1, 2$ where 1=urban residence and 2 = rural residence and $j = 1, 2, 3$ where 1=low, 2=medium, 3=high stress level

α = log odds of favorable response for urban residence and low stress levels

β_1 = increment in log odds for rural residence

β_2 = increment in log odds for medium stress levels

β_3 = increment in log odds for high stress levels

$$X_1 = \begin{cases} 1 & \text{residence} = \text{"rural"} \\ 0 & \text{residence} = \text{"urban"} \end{cases}$$

$$X_2 = \begin{cases} 1 & \text{stress level} = \text{"medium"} \\ 0 & \text{otherwise} \end{cases}$$

$$X_3 = \begin{cases} 1 & \text{stress level} = \text{"high"} \\ 0 & \text{otherwise} \end{cases}$$

b)

data response;

input residence med_stress high_stress resp count @@;

datalines;

```
0 0 0 1 50    0 0 0 0    15
0 1 0 1 85    0 1 0 0    68
0 0 1 1 102   0 0 1 0    87
1 0 0 1 35    1 0 0 0    20
1 1 0 1 77    1 1 0 0    68
```

```

1 0 1 1 90    1 0 1 0    64
;
proc print;
run;

proc logistic data = response;
    freq count;
    model resp(event='1') = residence med_stress high_stress / scale = none aggregate;
run;

```

Odds Ratio Estimates

Effect	Point Estimate	95% Wald Confidence Limits	
residence	0.957	0.716	1.280
med_stress	0.491	0.312	0.774
high_stress	0.523	0.335	0.819

The odds ratio for medium stress as compared to low stress is 0.491 with confidence limits (0.312,0.774) after controlling for residence. This means the odds of favorable response for participants with medium stress level are 0.491 times the odds of favorable response for those participants with low stress level after adjusting for residence.

c)

Analysis of Maximum Likelihood Estimates

Parameter	DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept	1	0.9071	0.2122	18.2698	<.0001
residence	1	-0.0436	0.1482	0.0868	0.7683
med_stress	1	-0.7109	0.2321	9.3805	0.0022
high_stress	1	-0.6473	0.2284	8.0303	0.0046

$95\% \text{ CI} = \exp(\hat{\beta} \pm z_{1-\alpha/2}(\text{SE}(\hat{\beta})))$
 $= \exp(-0.7109 \pm 1.96(0.2321)) = (0.312, 0.774)$

d)

$H_0 : \beta_1 = 0$

$H_a : \beta_1 \neq 0$

The Wald statistic is 0.0868 ($p = 0.7683$). The test statistic follows $N(0,1)$ under the null. We fail to reject the null since $p > 0.05$ and conclude that residence and opinion response are significantly associated at the 0.05 level.

e)

```

data health_policy;
input residence : $5. stress_level $ response $ count @@;
datalines;
urban low    F 50  urban low    U 15

```

```

urban medium F 85  urban medium U 68
urban high   F 102 urban high   U 87
rural low    F 35  rural low    U 20
rural medium F 77  rural medium U 68
rural high   F 90  rural high   U 64
;
proc print;
run;

proc logistic data = health_policy;
  freq count;
  class residence(ref='urban') stress_level(ref='low') / param=ref;
  model response(event='F') = residence stress_level;
  output out=predict pred=prob;
run;

proc print data = predict;
run;

```

Obs	residence	level	response	count	_LEVEL_	prob
1	urban	low	F	50	F	0.71241
2	urban	low	U	15	F	0.71241
3	urban	medium	F	85	F	0.54889
4	urban	medium	U	68	F	0.54889
5	urban	high	F	102	F	0.56460
6	urban	high	U	87	F	0.56460
7	rural	low	F	35	F	0.70338
8	rural	low	U	20	F	0.70338
9	rural	medium	F	77	F	0.53806
10	rural	medium	U	68	F	0.53806
11	rural	high	F	90	F	0.55384
12	rural	high	U	64	F	0.55384

Note that $\Pr(\text{unfavorable response}) = 1 - \Pr(\text{favorable response})$.

a. An individual from an urban area with low stress.

$\Pr(\text{unfavorable response}) = 1 - 0.71241 = 0.28759$

b. An individual from a rural area with high stress.

$\Pr(\text{unfavorable response}) = 1 - 0.55384 = 0.44616$

f) Chi-square goodness-of-fit tests will be used. We can use these tests because we have at least 10 subjects in each group. We have 80% predicted counts ≥ 5 and all other expected counts > 2 with no 0 counts.

Response Profile

Ordered		Total
Value	resp	Frequency

1	0	322
2	1	439

Probability modeled is resp=1.

Model Convergence Status

Convergence criterion (GCONV=1E-8) satisfied.

Deviance and Pearson Goodness-of-Fit Statistics

Criterion	Value	DF	Value/DF	Pr > ChiSq
Deviance	3.3269	2	1.6634	0.1895
Pearson	3.3246	2	1.6623	0.1897

Number of unique profiles: 6

If the model fits, Q_P and Q_L are distributed as χ^2 with $df = 6 \text{ rows} - 4 \text{ parameters} = 2$. The null hypothesis is the model fits the data and the alternative is the model does not fit the data. $Q_P = 3.3269$ ($p = 0.1895$) and $Q_L = 3.3246$ ($p = 0.1897$). Both of these p-values are > 0.05 so we would fail to reject the null and conclude that the model does indeed fit the data.

2.

a) The regression equation is:

$$\text{logit}(p) = \alpha + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3$$

where p = probability favorable outcome after a 1-year period and

α = log odds of favorable outcome for center 2 and poor baseline condition and placebo treatment

β_1 = increment in log odds for center 1

β_2 = increment in log odds for good baseline condition

β_3 = increment in log odds for test treatment

$$X_1 = \begin{cases} 1 & \text{Center} = 1 \\ 0 & \text{Center} = 2 \end{cases}$$

$$X_2 = \begin{cases} 1 & \text{baseline condition} = \text{"good"} \\ 0 & \text{baseline condition} = \text{"poor"} \end{cases}$$

$$X_3 = \begin{cases} 1 & \text{treatment} = \text{"test"} \\ 0 & \text{treatment} = \text{"placebo"} \end{cases}$$

b) The Wald Chi-Square is found by:

$$\left(\frac{\text{estimate}}{\text{SE}}\right)^2$$

Parameter	Estimate	Standard Error	Wald Chi-Square
Intercept	-1.056	0.255	17.15
Center (1 vs. 2)	-0.102	0.103	0.981
Baseline (good vs. poor)	0.826	0.324	6.5
Treatment (test vs. placebo)	3.132	1.020	9.429

c) 95% CI = $\exp(\hat{\beta} \pm z_{1-\alpha/2}(\text{SE}(\hat{\beta})))$

= $\exp(3.132 \pm 1.96(1.020)) = (3.104, 169.220)$

The confidence limits do not include 1, indicating that treatment is significantly associated with favorable outcome.

d) 95% CI = $\exp(\hat{\beta} \pm z_{1-\alpha/2}(\text{SE}(\hat{\beta})))$

= $\exp(0.826 \pm 1.96(0.324)) = (1.210, 4.310)$

The confidence limits do not include 1, indicating that baseline condition is significantly associated with favorable outcome.

e) Provide predicted probabilities of favorable outcome for these patients:

i) Center 1, poor baseline condition, and placebo treatment

$$\text{Pr}(\text{favorable outcome}) = \frac{e^{\alpha+\beta_1}}{1 + e^{\alpha+\beta_1}} = \frac{e^{-1.056-0.102}}{1 + e^{-1.056-0.102}} = 0.239$$

ii) Center 2, good baseline condition, and test treatment

$$\text{Pr}(\text{favorable outcome}) = \frac{e^{\alpha+\beta_2+\beta_3}}{1 + e^{\alpha+\beta_2+\beta_3}} = \frac{e^{-1.056+0.826+3.132}}{1 + e^{-1.056+0.826+3.132}} = 0.948$$

3.

Answering Q3

*****;

data dose;

input dose response \$ count @@;

datalines;

1 F 25 1 U 45

10 F 35 10 U 35

100 F 56 100 U 14

;

proc print;

run;

proc logistic data = dose;

freq count;

```

class dose(ref='1') / param = ref;
model response(event='F') = dose;
run;

```

a) The regression equation is:

$$\text{logit}(p) = \alpha + \beta_1 X_1 + \beta_2 X_2$$

where p = probability of favorable response

α = log odds of favorable outcome for center 2 and poor baseline condition and placebo treatment

β_1 = increment in log odds for 10 mg dose

β_2 = increment in log odds for 100 mg dose

$$X_1 = \begin{cases} 1 & \text{dose} = 10 \text{ mg} \\ 0 & \text{otherwise} \end{cases}$$

$$X_2 = \begin{cases} 1 & \text{dose} = 100 \text{ mg} \\ 0 & \text{otherwise} \end{cases}$$

Analysis of Maximum Likelihood Estimates

Parameter	DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept	1	-0.5878	0.2494	5.5526	0.0185
dose 10	1	0.5878	0.3455	2.8944	0.0889
dose 100	1	1.9741	0.3892	25.7211	<.0001

Odds Ratio Estimates

Effect	Point Estimate	95% Wald Confidence Limits
dose 10 vs 1	1.800	0.915 3.543
dose 100 vs 1	7.200	3.357 15.440

The odds of favorable response when dose is 1 mg are $\exp(-0.5878)=0.556$. The odds of favorable outcome for 10 mg dose are 1.80 times the odds of favorable outcome for the 1 mg dose. The odds of favorable outcome for the 100 mg dose are 7.20 times the odds of favorable outcome for the 1 mg dose. The odds ratio is found by exponentiating the parameter estimate from the above table.

b) The new regression equation is:

$$\text{logit}(p) = \alpha + \beta_1 X_1$$

where p is the probability of favorable response and β_1 = increment in log odds for one-unit increase in log of dose and X_1 = log of dose

```

data dose2;
  set dose;
  if missing(dose)=0 then log_dose = log10(dose);
run;

*New model;
proc logistic data = dose2 alpha=0.01;
  freq count;
  model response(event='F') = log_dose / lackfit;
run;

```

Analysis of Maximum Likelihood Estimates

Parameter	DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept	1	-0.7220	0.2318	9.6968	0.0018
log_dose	1	0.9646	0.1903	25.6893	<.0001

Odds Ratio Estimates

Effect	Point Estimate	99% Wald Confidence Limits
log_dose	2.624	1.607 4.284

As we have done a log transformation, the interpretation of the parameters changes. The interpretation of the intercept is the odds of favorable response when dose is 1 mg. If dose is 1 mg, then $\log_{10}(1) = 0$ so that term goes away in the model. Then the odds of favorable outcome when dose is 1 mg for the new model is $\exp(-0.7220)=0.486$. To interpret β_1 , we will do a back transformation. We can take $10^{0.9646} = 9.21$. This means that for every tenfold increase in dose, the odds of favorable outcome are 9.21 times more likely.

c) The MLE table for the new model is shown above in b). The OR estimate for $\log_{10}(\text{dose})$ is 2.624 with 99% confidence limits (1.607,4.284).

Deviance and Pearson Goodness-of-Fit Statistics

Criterion	Value	DF	Value/DF	Pr > ChiSq
Deviance	1.6852	1	1.6852	0.1942
Pearson	1.6852	1	1.6852	0.1942

Number of unique profiles: 3

If the model fits, Q_P is distributed as $\chi^2_{(1)}$. The null hypothesis is that the model fits the data. $Q_P = 1.6852$ ($p = 0.1942$). As this p-value is > 0.05 , we would fail to reject the null hypothesis and conclude that the model fit is adequate.